NONOS

Privacy Infrastructure Economy

A sustainable economic model where node operators earn rewards for delivering measurable privacy services —not for holding tokens. Work creates value. Value creates rewards.

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Introduction

Traditional staking models have failed privacy infrastructure. They reward passive capital holders while the actual operators running nodes, relaying traffic, and generating proofs receive minimal compensation. NONOS changes this with the Privacy Infrastructure Economy.

The Problem with Stake-Weighted Rewards

Most blockchain staking systems distribute rewards based on how many tokens a participant holds. A whale with millions of tokens earns proportionally more than a small operator, regardless of the actual work either performs. This creates several problems for privacy networks:

- **Misaligned incentives.** Large holders can earn substantial yields without running any infrastructure, while operators bearing real costs receive diminishing returns.
- **Centralization pressure.** Wealth concentrates among early investors and large funds, not among those building and maintaining the network.
- **Unsustainable economics.** When rewards flow to passive holders rather than active operators, the network depends on continuous token appreciation to attract infrastructure providers.

The Privacy Work Economy

NONOS introduces a fundamentally different model. Rewards flow to operators based on measurable privacy work performed, not tokens held. A node relaying terabytes of traffic earns more than a node sitting idle—regardless of how much each has staked.

Stake still exists, but only as **collateral against misbehavior**. Operators must stake a minimum amount to participate, but additional stake above the minimum provides zero reward benefit. This prevents plutocratic capture while maintaining economic security.

> **Core Principle:** A node with minimum collateral performing maximum work earns more than a node with massive stake performing minimal work. Work creates value. Value creates rewards.

The Privacy Liquidity Pool

At the center of the Privacy Infrastructure Economy sits the Privacy Liquidity Pool—a smart contract that collects fees from privacy services and distributes rewards to those who make privacy possible.

Pool Inflows

The pool receives funds from three sources, each serving a distinct economic purpose:

Privacy Service Fees

Every time someone uses NONOS privacy services—routing traffic through the anonymity network, generating a zero-knowledge proof, participating in a mixing round, or registering a stealth address—a small fee flows to the pool. These micro-fees accumulate continuously as the network processes privacy operations.

Bootstrap Emissions

During the first two years, the protocol emits 54,794 NOX daily from a dedicated bootstrap pool of 40 million tokens (5% of total supply). This subsidizes early operators while the network builds sufficient fee volume to sustain itself. The emission follows a fixed schedule and cannot be modified.

Liquidity Locks

Token holders who want to support network liquidity can lock NOX in the pool for fixed durations. Longer locks receive higher reward multipliers, incentivizing long-term commitment over short-term speculation.

Pool Outflows

Collected funds distribute to two recipient categories each epoch:

RECIPIENT	SHARE	BASIS
Node Operators	70%	Proportional to work score
Liquidity Providers	30%	Proportional to weighted lock amount

The 70/30 split is immutable, ensuring operators always receive the majority of rewards for performing actual work.

Measuring Privacy Work

For work-based rewards to function, the protocol must measure work accurately and resist gaming. NONOS tracks five categories of verifiable privacy contributions, each weighted according to its value to the network.

Work Categories

CATEGORY	WEIGHT	WHAT IT MEASURES
Traffic Relay	30%	Bandwidth relayed through anonymity circuits
ZK Proof Generation	25%	Zero-knowledge proofs computed for users
Mixer Operations	20%	Participation in mixing rounds
Entropy Provision	15%	Contributions to verifiable randomness
Registry Operations	10%	Stealth address registrations and lookups

Score Calculation

Each node's work score combines weighted contributions across all categories. To prevent gaming through cheap work inflation, raw metrics are normalized against network averages and capped at 5x the mean. A node cannot dominate rewards by flooding a single metric.

Epoch-Based Distribution

Work aggregates over seven-day epochs. At epoch end, oracles submit work scores and the protocol calculates each node's share of the reward pool:

> **Node Reward = (Node Work Score ÷ Total Network Work) × Node Pool**

A node performing 2% of total network work receives 2% of the node reward pool. Simple, transparent, and directly tied to contribution.

Collateral as Security

Stake in the Privacy Infrastructure Economy serves one purpose: economic security against malicious behavior. It does not influence reward calculations. Operators must meet minimum collateral requirements, but staking more provides no additional benefit.

Dynamic Minimum Requirements

The minimum collateral scales with network size. Early participants face lower barriers, while a mature network with thousands of nodes requires more substantial commitment. This balances accessibility during growth with security at scale.

The formula applies a square root function to network size, creating sublinear growth that remains accessible even as the network expands:

> **Minimum Collateral = 10,000 NOX × √(Active Nodes ÷ 10)**

NETWORK SIZE	MINIMUM STAKE
10 nodes	10,000 NOX
100 nodes	31,623 NOX
1,000 nodes	100,000 NOX
10,000 nodes	316,228 NOX

Slashing for Misbehavior

Collateral can be slashed for verifiable protocol violations. False work claims result in 5% slashing. Privacy violations—such as logging user data or performing traffic correlation—trigger 10% slashing. Active attacks like Sybil or eclipse attempts incur 25% penalties. Complete key compromise results in full collateral loss.

All slashing requires cryptographic proof and passes through a seven-day challenge period where operators can dispute claims. Slashed funds split between the reporter who identified the violation and the protocol treasury.

Liquidity Provider Rewards

Token holders who do not operate nodes can still participate by locking liquidity in the pool. Thirty percent of pool rewards flow to liquidity providers, with longer lock durations receiving higher multipliers.

Lock Duration Tiers

The protocol offers five lock durations. Each tier carries a multiplier that increases effective weight in the reward calculation:

DURATION	MULTIPLIER	EFFECTIVE BOOST
14 days	1.00×	Baseline
30 days	1.25×	+25%
90 days	1.60×	+60%
180 days	2.00×	+100%
365 days	2.50×	+150%

A provider locking 100,000 NOX for one year has effective weight of 250,000 in reward calculations, compared to 100,000 for a two-week lock of the same amount.

Early Exit Penalties

Providers who withdraw before their lock expires pay penalties that redistribute to remaining participants. Exiting within the first quarter of a lock incurs 50% penalty. Between 25-50%, the penalty is 30%. Between 50-75%, it drops to 15%. In the final quarter, only 5% applies.

This structure rewards commitment while allowing emergency exits at a cost. Eighty percent of penalty funds redistribute to other liquidity providers; twenty percent flows to the protocol treasury.

Privacy Service Fees

Protocol fees create sustainable economics beyond the two-year bootstrap period. Each privacy service carries a small fee that flows to the pool, aligning user costs with infrastructure rewards.

Service Fee Schedule

SERVICE	FEE	DESCRIPTION
Priority Relay	10 NOX/GB	Fast-track circuit routing
ZK Proof (Standard)	5 NOX	Groth16 proof generation
ZK Proof (Complex)	25 NOX	PLONK/STARK proof generation
Standard Mixing	0.1%	Percentage of mix amount
Priority Mixing	0.25%	Immediate mixing execution
Stealth Registration	50 NOX	One-time address registration
Stealth Lookup	1 NOX	Address resolution query

Enterprise API Access

Applications and services integrating NONOS privacy features can subscribe to API tiers providing higher rate limits and dedicated support:

- **Basic (1,000 NOX/month):** 1,000 requests per day, standard APIs
- **Pro (5,000 NOX/month):** 10,000 requests per day, priority support
- **Enterprise (25,000 NOX/month):** Unlimited requests, dedicated nodes

Fee Flow

All fees collect in the Fee Router contract, which flushes accumulated amounts to the Privacy Liquidity Pool at epoch boundaries. Fees follow the same 70/30 distribution as emissions—seventy percent to node operators based on work, thirty percent to liquidity providers based on weighted locks.

Bootstrap to Sustainability

The Privacy Infrastructure Economy transitions from subsidized bootstrap to fee-sustained operations over 730 days. This section details the emission schedule and sustainability requirements.

Token Economics

800M	40M	730
Total NOX Supply	Bootstrap Pool (5%)	Distribution Days

The bootstrap pool of 40 million NOX distributes linearly over two years at 54,794.52 NOX per day. Combined with fees, this creates weekly epoch pools of approximately 383,561 NOX plus accumulated fees.

Post-Bootstrap Economics

After day 730, emissions stop permanently. The protocol operates entirely on fee revenue. For the network to maintain current reward levels, it must generate approximately **55,000 NOX in weekly fees**—equivalent to the bootstrap emission rate.

At current fee rates, this requires meaningful usage volume: roughly 5,500 GB of priority relay traffic, 11,000 standard ZK proofs, or equivalent combinations of services. The two-year bootstrap period provides time to reach these thresholds.

No Inflation

Total supply is fixed at 800 million NOX. The protocol cannot mint additional tokens to extend emissions or subsidize operations. This creates hard constraints that force genuine value creation—the network must provide services people will pay for, or rewards decline.

> **Sustainability Target:** The network must generate sufficient fee volume by day 730 to maintain operator incentives. Governance may propose treasury subsidies as a temporary bridge, but cannot create new tokens.

Oracle Network & Security

Work-based rewards require accurate, manipulation-resistant work measurement. The protocol relies on a decentralized oracle network to aggregate and submit work scores.

Oracle Consensus

Five independent oracle operators measure node work throughout each epoch. At epoch end, oracles submit their observed work scores. The protocol requires **three of five** oracles to agree before accepting a submission, preventing any single operator from manipulating scores.

Oracle operators must stake 100,000 NOX as collateral and face slashing for provably false submissions. Quarterly rotation of the oracle set prevents long-term collusion.

Attack Resistance

The system incorporates multiple defenses against economic attacks:

- **Sybil resistance:** Collateral requirements make spinning up fake nodes expensive. Work verification ensures nodes actually perform services.
- **Work gaming prevention:** Score normalization and caps prevent flooding cheap metrics. Cross-category requirements ensure balanced contribution.
- **Flash loan immunity:** Epoch-based rewards with one-epoch claim delays prevent same-block manipulation.
- **Oracle manipulation:** Consensus requirements, stake slashing, and rotation limit oracle attack surface.

Circuit Breakers

Automated safety mechanisms trigger under anomalous conditions. If total value locked drops 50% within 24 hours, withdrawals pause pending multisig review. Anomalous work score patterns delay epoch finalization for oracle investigation. These safeguards provide time to respond to attacks without requiring continuous monitoring.

Governance

The Privacy Infrastructure Economy balances immutable commitments with adaptable parameters. Core economics cannot change; operational details can evolve through governance.

Immutable Parameters

Certain values are hardcoded and cannot be modified by any governance process:

- Total supply: 800 million NOX
- Bootstrap pool: 40 million NOX (5%)
- Distribution period: 730 days
- Node/LP reward split: 70/30
- Work category weights: 30/25/20/15/10
- Epoch duration: 7 days

These constraints ensure participants can trust the fundamental economics regardless of future governance decisions.

Adjustable Parameters

Secondary parameters may be modified through governance proposals with appropriate timelocks:

- Lock duration multipliers (within 1.0× to 3.0× bounds)
- Base stake requirement (within 5,000 to 50,000 NOX)
- Slashing percentages (within defined ranges)
- Early exit penalty rates
- Service fee amounts
- New work category additions

Proposal Process

Creating a proposal requires holding 100,000 NOX. Voting runs for seven days with 4% quorum requirement. Passed proposals enter a timelock of 7-30 days before execution, providing time for participants to react to changes they oppose.

Emergency functions—pause, unpause, and emergency withdrawal—require three-of-five multisig approval and bypass normal governance for rapid response to critical situations.

Implementation Components

Bringing the Privacy Infrastructure Economy to life requires coordinated development across smart contracts, oracle infrastructure, daemon integration, and user interfaces.

Smart Contracts

Core Contracts

- **Privacy Liquidity Pool** — Central pool holding emissions and fees
- **Work Registry** — Stores oracle-submitted work scores
- **Reward Distributor** — Calculates and distributes epoch rewards

Supporting Contracts

- **Collateral Manager** — Handles stakes and slashing
- **Lock Manager** — LP locks with tier multipliers
- **Fee Router** — Collects and routes service fees

Oracle Infrastructure

The Work Oracle network requires five independent operators running measurement nodes. These nodes connect to NONOS daemon instances across the network, aggregate work metrics, and submit consensus scores on-chain at epoch boundaries.

Daemon Integration

The NONOS daemon must report work metrics to the oracle network. This includes bandwidth counters, proof generation logs, mixing participation records, entropy contributions, and registry operation counts. Metrics transmit through privacy-preserving aggregation to prevent deanonymization.

User Interface

The NONOS application exposes economy features through dedicated screens:

- **Node Dashboard** — Work scores, earnings history, collateral status
- **Liquidity Panel** — Lock management, tier selection, reward claims
- **Service Usage** — Fee estimates, usage history, API management

Conclusion

The Privacy Infrastructure Economy represents a fundamental rethinking of how privacy networks sustain themselves. By rewarding work instead of wealth, NONOS creates aligned incentives for building robust, decentralized privacy infrastructure.

Key Innovations

- **Work-based rewards** ensure those providing actual privacy services receive compensation proportional to their contribution.
- **Stake as collateral** maintains economic security without creating plutocratic reward concentration.
- **Additive fee model** builds sustainable economics that can operate indefinitely after bootstrap emissions end.
- **Tiered liquidity locks** reward long-term commitment while providing flexibility for participants.
- **Immutable core parameters** provide trust guarantees that governance cannot undermine fundamental economics.

The Path Forward

Privacy is not a luxury—it is infrastructure. The internet was built on the assumption that privacy would be someone else's problem. NONOS provides the economic foundation for privacy infrastructure that can sustain itself through genuine value creation.

Node operators who relay traffic, generate proofs, and facilitate mixing create real value for users seeking privacy. The Privacy Infrastructure Economy ensures that value flows back to those who create it.

> **Work creates value. Value creates rewards.** This is the foundation of the Privacy Infrastructure Economy.

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